

Lab 2 — Measuring a Wavelength You Cannot See

SYNOPSIS

In Lab 1 you measured an oscillation that sat still on a bench, and found its frequency and amplitude with a stopwatch and a ruler. This lab measures something you cannot see, cannot touch, and cannot hold still: the **wavelength of a sound wave** crossing the room. The trick is the one you met in Note 4a. Two speakers playing the same tone produce places where the sound is loud and places where it nearly vanishes, and those places are fixed in space by one condition — the difference in the distances from the two speakers, measured in wavelengths. Locate a quiet spot, measure two distances with a tape, subtract, and you have measured half a wavelength of something moving at 343 metres per second. Then you turn it around: from λ and $v = \lambda f$ you can compute the frequency of the tone — and unlike almost every measurement you will make this semester, **you already know the right answer**, because the tone was dialed in. That makes this lab a test of your method as much as a measurement, exactly as the calibrated simulation was in Lab 1.

Performance Expectations: Locate interference minima in a real room by ear and show by repeat measurement that the locations are reproducible. Measure two path lengths and obtain a wavelength from their difference using $\Delta = (n + \frac{1}{2})\lambda$. Compute a frequency from the measured wavelength using $v = \lambda f$ and compare it against the known dialed frequency as a percentage error. Predict and then test how the pattern changes when the frequency is changed. Perform the same measurement in a simulation that has no reflections, and use the comparison to identify and explain the sources of error in the real-room measurement. Distinguish random error (spread between trials) from systematic error (a consistent offset), citing evidence from your own data.

Materials: two speakers driven from one amplifier or one audio source; a laptop or phone running a tone generator; a tape measure (5 m or longer); masking tape to mark the floor; a phone with a sound-level app (phyphox, “Audio Amplitude”) — optional but recommended; a computer or phone running the PhET *Wave Interference* simulation.

BEFORE YOU START

Both speakers must play the **same** signal — one source, one amplifier. Stand at the midpoint between them and confirm it is **loud** there. If the midpoint is quiet, one speaker is wired backwards; tell your instructor before you take any data, because every number in Part 1 will be shifted otherwise.

Part 1 — The real room

Setup

Place the two speakers on the bench at the same height, facing the same direction, parallel — not angled toward each other. Measure the distance **d** between the two speaker cones and record it. Run a line of masking tape across the floor parallel to the speakers, roughly **L** = 3 m out, and mark the point on it directly opposite the midpoint of the two speakers: this is your **centre mark**, position 0. Set the tone generator to the frequency your instructor gives you and record it. Take the speed of sound as $v = 343 \text{ m/s}$.

Speaker separation d (m)	Walk-line distance L (m)	Dialed frequency f (Hz)	Speed of sound v (m/s)	Expected $\lambda = v/f$ (m)
			343	

COMPUTE THE EXPECTED λ NOW, BEFORE YOU MEASURE ANYTHING

Writing down what you expect *before* you take data is what makes the comparison afterwards meaningful. It is also the only way to catch a measurement that is wrong by a factor of two — the most common failure in this lab — while you are still standing next to the apparatus.

A. Find the quiet spots

- 1 Plug **one ear**. Keep your head in the same orientation for the whole walk. Start at the centre mark and walk slowly along the tape to one side, listening for the tone to fade.
- 2 When you find the faintest point, stop and have your partner put a small piece of tape on the floor at your feet. Keep walking — find the next loud spot, then the **second** quiet spot, and mark that too.
- 3 Measure each mark's distance from the centre mark along the tape, and record it as trial 1.
- 4 **Swap people** and repeat the whole walk twice more, each time with a different person listening, marking independently. Do not let the second and third walkers look at the earlier marks before they commit.

Compute the average position of each null and the **spread** (largest minus smallest) across your three trials.

	Trial 1 (m)	Trial 2 (m)	Trial 3 (m)	Average (m)	Spread (m)
First quiet spot from centre					
Second quiet spot from centre					

That spread column is your **random error**, and it is data, not an embarrassment. Three people disagreeing by 5 cm tells you something honest about how sharply a null can be located by ear. Three people disagreeing by 50 cm tells you something else, and you should find out what before continuing.

B. Turn two distances into a wavelength

- 1 Stand at the **average** first-null position. Measure the distance from that point to each speaker cone with the tape. Record r_A and r_B , then take $|r_A - r_B|$.
- 2 Repeat at the **average** second-null position.
- 3 For the first null, $n = 0$, so $\Delta = \frac{1}{2}\lambda$ and therefore $\lambda = 2\Delta$. For the second null, $n = 1$, so $\Delta = \frac{3}{2}\lambda$ and $\lambda = \frac{2\Delta}{3}$. Compute λ from each null separately.
- 4 Average your two values of λ , and compare with the expected λ from Table 0 as a percentage difference.

Null	n	r_A (m)	r_B (m)	Δ (m)	λ from this null (m)
first	0				$\lambda = 2\Delta =$
second	1				$\lambda = 2\Delta/3 =$
Average λ					
Expected λ (Table 0)					
Percentage difference					

$$\% \text{ difference} = |\text{measured} - \text{expected}| \div \text{expected} \times 100\%$$

Use this same form everywhere in your report. **Always divide by the value you trust** — here, the expected one.

C. Now measure the frequency you were given

Your average λ came from a tape measure and nothing else. Combine it with the speed of sound to recover the frequency, then compare with the number that was dialed into the generator.

1

Compute $f = v / \lambda_{\text{measured}}$.

$f_{\text{measured}} =$ _____ Hz

2

Compare with the dialed frequency as a percentage difference.

% difference = _____

D. Change the frequency

- 1

Predict first, in writing: your instructor will raise the tone to roughly double the frequency. Before anything is changed, state what will happen to λ , and what will happen to the distance from the centre mark to the first quiet spot.

λ will: _____

first-null distance will: _____
- 2

With the new frequency running, repeat section A once (one walk is enough) and find the first quiet spot. Then repeat section B at that null to obtain a new λ .
- 3

Fill in the comparison table. Was your prediction right?

	Dialed f (Hz)	Expected $\lambda = v/f$ (m)	Measured first-null distance (m)	Measured λ (m)
original tone				
higher tone				

Part 2 — The same measurement with the room removed

Your Part 1 numbers carry the room in them: sound bouncing off the walls, the floor, the benches and the other students arrives at your ear along with the two direct paths, and partly fills in the nulls. The simulation has none of that. It is the same experiment with every source of error stripped out, which makes it the standard you judge Part 1 against — the same role the calibrated oscillator played in Lab 1.

Open the PhET *Wave Interference* simulation and choose the **Interference** panel, then switch it to **Sound**.

- 1

Method 1 — measure λ directly. With one speaker running, use the measuring tape to find the distance from one crest to the next.

λ_{direct} = _____ cm

2

Method 2 — measure λ the way you did in Part 1. Turn on both speakers. Put a probe in the middle of a quiet band, measure the distance from it to each speaker, subtract, and convert the path difference into a wavelength using the correct n for the band you chose.

$r_A =$ _____ $r_B =$ _____ $\Delta =$
 _____ cm

$n =$ _____ $\lambda_{\text{path}} =$ _____ cm

3

Compare the two methods as a percentage difference.

% difference = _____

4

Repeat step 2 at a **different** quiet band, with a different n , and confirm you recover the same λ . If you do not, the most likely cause is that you have the wrong n — count the bands outward from the centre line carefully.

second band: $n =$ _____ **$\lambda =$** _____ cm

	Real room (Part 1)	Simulation (Part 2)
% difference between methods / from expected		
Spread between repeat trials		
How sharply defined was the quiet spot?		

Questions

1

State in your own words why a quiet spot exists at all. Your answer must use the words *path*, *difference* and *phase*, and must not use the word “cancel” without explaining what cancels what.

2

At the first quiet spot you set $n = 0$ and used $\lambda = 2\Delta$. Explain where the factor of 2 comes from. What would you have computed if you had wrongly assumed the first quiet spot was $n = 1$ — and by what factor would your λ have been off?

3

The midpoint between the speakers is loud no matter what frequency is playing. Explain why, in terms of Δ , without arithmetic.

4

Why did you take three independent walks in section A instead of trusting one? What specifically did the **spread** column tell you that a single trial could not?

5

Your measured λ differs from the expected λ . Classify each of the following as a source of **random** error or **systematic** error, and say which direction, if any, a systematic one would push your result:

- (a) different people hearing the null at slightly different places;
- (b) sound reflecting off the back wall;
- (c) measuring to the front of the speaker box rather than to the cone;
- (d) the actual air temperature differing from the value assumed in $v = 343$ m/s.

6

In Part 1C you computed a frequency from a tape measure and compared it with a number you already knew. What is the point of measuring something whose answer you have been told? What does a small percentage difference license you to claim about the rest of your data?

7

The quiet spots in the room were never *silent*, but in the simulation the calm bands are much cleaner. Name the thing the real room has that the simulation does not, and explain how it fills in a null.

8

From your Part 1D data: when the frequency roughly doubled, what happened to the wavelength and to the null spacing? Express the relationship between f and λ as a proportionality, and identify what was held constant to make it true.

9

Suppose you repeated this lab outdoors on a cold morning at $-5\text{ }^{\circ}\text{C}$, where the speed of sound is about 328 m/s rather than 343 m/s. Using the same dialed frequency, would the quiet spots be closer together or farther apart than they were indoors? Justify from $v = \lambda f$ and $\Delta = (n + \frac{1}{2})\lambda$.

10

You have now measured the wavelength of something you cannot see, using only distances. Describe in two or three sentences how you would use the same method to measure the wavelength of *light*, and name the one practical difficulty that makes it much harder.

LAB REPORT

Follow the standard format in the **Writing Lab Reports** module — abstract, data tables, and all required sections. This handout will not remind you of every part. Your abstract must state your measured λ , your measured f , and the percentage difference from the dialed value, in that order, in one sentence.

Files for this lab
